

Machine Learning - Answer Key

Question (20/25 Marks): Explain Ensemble Learning in detail.

Ensemble Learning is a machine learning technique where multiple models (often called weak learners)

are combined to produce a stronger and more robust model. The main idea is that a group of weak learners

can come together to form a strong learner, improving overall performance, accuracy, and generalization.

There are three main types of Ensemble Learning methods:

1. Bagging (Bootstrap Aggregating):

Bagging reduces variance by training multiple models on different subsets of the dataset.

Each subset is created using random sampling with replacement.

Example: Random Forest, where multiple decision trees are trained and their outputs are averaged.

2. Boosting:

Boosting focuses on reducing bias by training models sequentially. Each new model gives more importance

to previously misclassified data points.

Examples:

- AdaBoost (Adaptive Boosting)
- Gradient Boosting
- XGBoost

3. Stacking:

Stacking combines multiple different models and uses another model (meta-learner) to learn how to best

combine their predictions.

Advantages of Ensemble Learning:

- Improves accuracy and performance
- Reduces overfitting (especially in bagging)
- Handles complex data better
- Provides better generalization

Disadvantages:

- Computationally expensive
- Harder to interpret compared to single models
- Requires more memory and time

Applications:

- Image classification
- Fraud detection
- Medical diagnosis
- Recommendation systems

Conclusion:

Ensemble Learning is a powerful approach in machine learning that combines multiple models to achieve

better predictive performance compared to individual models.